

Computational Methods in Finance

Lecture – Week 12

Financial Engineering

Industrial Engineering & Operations Research

Columbia University

For educational purposes only, based on the book titled Computational Methods in Finance by Ali Hirsra

Agenda

- Kalman Filter (KF)
- Extended Kalman Filter (EKF)
- Unscented Kalman Filter (UKF)
- Square-Root Unscented Kalman Filter (SR-UKF)
- Particle Filtering (PF)

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Kalman Filter

filtering is a tool for tracking the state of a dynamic system modeled by some network

in KF, we assume the following linear model for the evolution of the true state at time t_k :

$$x_k = F_k x_{k-1} + B_k u_k + w_k$$

where

- ▷ x_k is the (true) state at time t_k
- ▷ F_k is the state transition matrix
- ▷ B_k is the control-input model applied to the control vector u_k
- ▷ w_k is the process noise assumed to be a multivariate normal distribution with zero mean and covariance Q_k , that is

$$w_k \sim \mathcal{N}(0, Q_k)$$

▷ or

$$Q_k = \mathbb{E}(w_k w_k^\top)$$

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Kalman Filter

- x_k is an $n \times 1$ vector of real numbers and F_k is an $m \times n$ matrix
- at time k an observation (measurement) z_k of state x_k is made and we assume the following measurement equation for its evolution:

$$z_k = H_k x_k + v_k$$

- where H_k is the observation matrix and v_k is the observation noise assumed to be Gaussian with zero mean and covariance R_k , that is,

$$v_k \sim \mathcal{N}(0, R_k)$$

or

$$R_k = \mathbb{E}(v_k v_k^\top)$$

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Kalman Filter

- initial state x_0 and the state noise vectors $w_1, \dots, w_k$ and measurement noise vectors $v_1, \dots, v_k$ are assumed to be mutually independent
- there are rare cases in which a dynamical system follows the above framework
- nonetheless considering the Kalman filter is designed to operate in the presence of noise makes it a useful filter and should be a good start
- we later describe variations and extensions on the Kalman filter that would allow more sophisticated models

Kalman Filter

- state of the filter is represented by the following variables:
 - $\hat{x}_{k|k-1}$: estimate of the state at time k given observations up to and including time $k-1$
 - $\hat{x}_{k|k}$: estimate of the state at time k given observations up to and including time k
 - $P_{k|k-1}$: error covariance matrix (a measure of the estimated accuracy of the state estimate) at time k given observations up to and including time $k-1$, that is

$$P_{k|k-1} = \mathbb{E}((x_k - \hat{x}_{k|k-1})(x_k - \hat{x}_{k|k-1})^\top)$$

- $P_{k|k}$: error covariance matrix at time k given observations up to and including time k , that is

$$P_{k|k} = \mathbb{E}((x_k - \hat{x}_{k|k})(x_k - \hat{x}_{k|k})^\top)$$

Measurement update

if the model is accurate and values for $\hat{x}_{0|0}$ and $P_{0|0}$ accurately reflect the distribution of the initial state values, then the following invariants are preserved:

- (a) all estimates have mean error zero

$$\begin{aligned}\mathbb{E}(x_k - \hat{x}_{k|k}) &= 0 \\ \mathbb{E}(x_k - \hat{x}_{k|k-1}) &= 0 \\ \mathbb{E}(z_k - H_k \hat{x}_{k|k-1}) &= 0\end{aligned}$$

- (b) covariance matrices accurately reflect the covariance of estimates

$$\begin{aligned}P_{k|k} &= \text{cov}(x_k - \hat{x}_{k|k}) \\ &= \mathbb{E}((x_k - \hat{x}_{k|k-1})(x_k - \hat{x}_{k|k-1})^\top) \\ P_{k|k-1} &= \text{cov}(x_k - \hat{x}_{k|k-1}) \\ &= \mathbb{E}((x_k - \hat{x}_{k|k})(x_k - \hat{x}_{k|k})^\top) \\ S_k &= \text{cov}(z_k - H_k \hat{x}_{k|k-1}) \\ &= \mathbb{E}((z_k - H_k \hat{x}_{k|k-1})(z_k - H_k \hat{x}_{k|k-1})^\top)\end{aligned}$$

Kalman Filter

- filtering is done in two steps:
 - (a) *time update or prediction*
 - &
 - (b) *measurement update or filtering*
- in the prediction phase, we use the state estimate from the previous time step in order to produce an estimate of the state at the current time step
- in the filtering phase, we use the measurement at the current time step to refine this prediction to obtain a new and (hopefully) a more accurate state estimate for the current time step

Time update

predicted state estimate:

$$\hat{x}_{k|k-1} = F_k \hat{x}_{k-1|k-1} + B_{k-1} u_{k-1}$$

predicted estimate covariance¹:

$$\begin{aligned}
 P_{k|k-1} &= \mathbb{E} \left((x_k - \hat{x}_{k|k-1})(x_k - \hat{x}_{k|k-1})^T \right) \\
 &= \mathbb{E} \left((F_k x_{k-1} + w_{k-1} - F_k \hat{x}_{k-1|k-1})(F_k x_{k-1} + w_{k-1} - F_k \hat{x}_{k-1|k-1})^T \right) \\
 &= \mathbb{E} \left((F_k (x_{k-1} - \hat{x}_{k-1|k-1}) + w_{k-1})(F_k (x_{k-1} - \hat{x}_{k-1|k-1}) + w_{k-1})^T \right) \\
 &= F_k \mathbb{E} \left((x_{k-1} - \hat{x}_{k-1|k-1})(x_{k-1} - \hat{x}_{k-1|k-1})^T \right) F_k^T + \mathbb{E} (w_{k-1} w_{k-1}^T) \\
 &= F_k P_{k-1|k-1} F_k^T + Q_{k-1}
 \end{aligned}$$

¹note in the derivation we assume the measurement error w_{k-1} is uncorrelated with the other terms

Measurement update

measurement residual:

$$\hat{\delta}_k = z_k - H_k \hat{x}_{k|k-1}$$

residual covariance²:

$$\begin{aligned} S_k &= \mathbb{E}(\hat{\delta}_k \hat{\delta}_k^\top) \\ &= \mathbb{E}((z_k - H_k \hat{x}_{k|k-1})(z_k - H_k \hat{x}_{k|k-1})^\top) \\ &= \mathbb{E}((H_k x_{k|k-1} + v_k - H_k \hat{x}_{k|k-1})(H_k x_{k|k-1} + v_k - H_k \hat{x}_{k|k-1})^\top) \\ &= \mathbb{E}((H_k(x_{k|k-1} - \hat{x}_{k|k-1}) + v_k)(H_k(x_{k|k-1} - \hat{x}_{k|k-1}) + v_k)^\top) \\ &= H_k \mathbb{E}((x_{k|k-1} - \hat{x}_{k|k-1})(x_{k|k-1} - \hat{x}_{k|k-1})^\top) H_k^\top + \mathbb{E}(v_k v_k^\top) \\ &= H_k P_{k|k-1} H_k^\top + R_k \end{aligned}$$

²note in the derivation we assume the measurement error v_k is uncorrelated with the other terms

Measurement update

updated state estimate³:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \hat{\delta}_k$$

updated estimate covariance⁴:

$$\begin{aligned}
 P_{k|k} &= \mathbb{E} \left((x_k - \hat{x}_{k|k})(x_k - \hat{x}_{k|k})^\top \right) \\
 &= \mathbb{E} \left((x_k - (\hat{x}_{k|k-1} + K_k \hat{\delta}_k))(x_k - (\hat{x}_{k|k-1} + K_k \hat{\delta}_k))^\top \right) \\
 &= \mathbb{E} \left((x_k - (\hat{x}_{k|k-1} + K_k(z_k - H_k \hat{x}_{k|k-1}))) (x_k - (\hat{x}_{k|k-1} + K_k(z_k - H_k \hat{x}_{k|k-1})))^\top \right) \\
 &= \mathbb{E} \left((x_k - (\hat{x}_{k|k-1} + K_k(H_k x_k + v_k - H_k \hat{x}_{k|k-1}))) (x_k - (\hat{x}_{k|k-1} + K_k(H_k x_k + v_k - H_k \hat{x}_{k|k-1})))^\top \right) \\
 &= \mathbb{E} \left(((I - K_k H_k)(x_k - \hat{x}_{k|k-1}) - K_k v_k) ((I - K_k H_k)(x_k - \hat{x}_{k|k-1}) - K_k v_k)^\top \right) \\
 &= \mathbb{E} \left(((I - K_k H_k)(x_k - \hat{x}_{k|k-1})) ((I - K_k H_k)(x_k - \hat{x}_{k|k-1}))^\top \right) + \mathbb{E} \left((K_k v_k) (K_k v_k)^\top \right) \\
 &= (I - K_k H_k) \mathbb{E} \left((x_k - \hat{x}_{k|k-1})(x_k - \hat{x}_{k|k-1})^\top \right) (I - K_k H_k)^\top + K_k \mathbb{E} \left(v_k v_k^\top \right) K_k^\top \\
 &= (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^\top + K_k R_k K_k^\top
 \end{aligned}$$

³ K_k is the Kalman gain (we do not make any assumption yet on its optimality)

⁴note in the derivation we assume the measurement error v_k is uncorrelated with the other terms and also use the definitions of $P_{k|k-1}$ and R_k

Measurement update

from the previous slide we have

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^T + K_k R_k K_k^T$$

and it holds for any arbitrary value for K_k

so far we have not made any assumption on K_k . We can reduce this expression further for the optimal Kalman gain

Posterior Estimate Covariance under Optimal Kalman Gain and Interpretation of the Optimal Kalman Gain

the Kalman filter minimizes posterior state estimation which is equivalent to saying it minimizes mean-square error estimator

the error in the posterior state estimation is

$$\epsilon_k \equiv x_k - \hat{x}_{k|k}$$

the goal is to minimize the expected value of the square of ϵ_k

$$\mathbb{E}(|x_k - \hat{x}_{k|k}|^2)$$

which is the same as minimizing the trace of the posterior estimate covariance matrix $P_{k|k}$.

Posterior Estimate Covariance

we get the following after expanding the terms in $P_{k|k}$:

$$\begin{aligned} P_{k|k} &= P_{k|k-1} - K_k H_k P_{k|k-1} - P_{k|k-1} H_k^\top K_k^\top + K_k (H_k P_{k|k-1} H_k^\top + R_k) K_k^\top \\ &= P_{k|k-1} - K_k H_k P_{k|k-1} - P_{k|k-1} H_k^\top K_k^\top + K_k S_k K_k^\top \end{aligned}$$

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Posterior Estimate Covariance

taking first derivative of the trace of $P_{k|k}$ with respect to K_k and setting it equal to zero, find optimal k_k :

$$\frac{\partial \text{tr}(P_{k|k})}{\partial K_k} = -2(H_k P_{k|k-1})^\top + 2K_k S_k = 0$$

solving it for K_k yields

$$K_k S_k = (H_k P_{k|k-1})^\top = P_{k|k-1} H_k^\top$$

or

$$K_k = P_{k|k-1} H_k^\top S_k^{-1}$$

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Posterior Estimate Covariance

this Kalman gain minimizes the mean-square error estimate and is the optimal Kalman gain.

$$K_k S_k K_k^\top = P_{k|k-1} H_k^\top K_k^\top$$

substituting back into $P_{k|k}$ yields

$$P_{k|k} = P_{k|k-1} - K_k H_k P_{k|k-1} - \cancel{P_{k|k-1} H_k^\top K_k^\top} + \cancel{P_{k|k-1} H_k^\top K_k^\top}$$

noticing that the last two terms cancel out give us

$$\begin{aligned} P_{k|k} &= P_{k|k-1} - K_k H_k P_{k|k-1} \\ &= (I - K_k H_k) P_{k|k-1} \end{aligned}$$

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Posterior Estimate Covariance

an interpretation of the Kalman filter could be based on linear regression

in the case of having time series of $\{\mathbf{x}_k\}$ and $\{\mathbf{z}_k\}$ the linear regression yields

$$\mathbf{z}_k = \alpha + \beta \mathbf{x}_k + \epsilon_k$$

with α the intercept, β the slope, and ϵ_k the residual

under linear regression we have

$$\beta = P_{k|k-1} H_k^\top S_k^{-1}$$

which is the expression for the Kalman gain.

Posterior Estimate

in Kalman filter, log likelihood for each time step is

$$\log p(z_t | z_{1:t-1})$$

which is obtained by evaluating the log of the probability density function of a multivariate Gaussian density with mean zero and covariance of S_k evaluated at the values in $\hat{\delta}_k$, that is, the log likelihood of innovation

$$\begin{aligned} \log p(z_t | z_{1:t-1}) &= \log \left(\frac{1}{(2\pi)^{d/2} |S_k|^{1/2}} \exp \left(-\frac{1}{2} (\hat{\delta}_k - 0)^\top S_k^{-1} (\hat{\delta}_k - 0) \right) \right) \\ &= -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |S_k| - \frac{1}{2} (\hat{\delta}_k - 0)^\top S_k^{-1} (\hat{\delta}_k - 0) \\ &= -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |S_k| - \frac{1}{2} (z_k - H_k \hat{x}_{k|k-1})^\top S_k^{-1} (z_k - H_k \hat{x}_{k|k-1}) \end{aligned}$$

which means we just need to focus on

$$-\log |S_k| - (z_k - H_k \hat{x}_{k|k-1})^\top S_k^{-1} (z_k - H_k \hat{x}_{k|k-1}) \quad (1)$$

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Posterior Estimate

- calling this part of log likelihood $\ln(\ell_k)$, then the log likelihood for the entire time series is $\ln(L_{1:T}) = \sum_{k=1}^T \ln(\ell_k)$
- having the log likelihood, we can either use an optimization routine (e.g., Nelder–Mead simplex) or employ the EM algorithm to minimize the negative log likelihood for parameter estimation
- it is worth mentioning that the Kalman gain does not come into the parameter estimation; it only enters into the filtering step

Extended Kalman Filter

- as in the Kalman filter, we call predicted state or a priori process estimates $\hat{x}_{k|k-1}$ and updated state estimate or a posterior estimate $\hat{x}_{k|k}$
- starting with our invariant on the error covariance $P_{k|k}$ as above

$$P_{k|k-1} = \text{cov}(x_k - \hat{x}_{k|k-1})$$

$$P_{k|k} = \text{cov}(x_k - \hat{x}_{k|k})$$

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Extended Kalman Filter

- matrix of partial derivatives, the Jacobians, are formed as follows:

$$F_{ij} = \frac{\partial f_i}{\partial x_j}(\hat{x}_{k-1|k-1}, 0)$$

$$U_{ij} = \frac{\partial f_i}{\partial u_j}(\hat{x}_{k-1|k-1}, 0)$$

$$H_{ij} = \frac{\partial h_i}{\partial x_j}(\hat{x}_{k|k-1}, 0)$$

$$V_{ij} = \frac{\partial h_i}{\partial v_j}(\hat{x}_{k|k-1}, 0)$$

- F_{ij} and U_{ij} are evaluated at $\hat{x}_{k-1|k-1}$ with zero noise and H_{ij} and V_{ij} are evaluated at $\hat{x}_{k|k-1}$ with zero noise

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Extended Kalman Filter

- having the Jacobians, we follow what was done in the Kalman filter to write the time update equation. For the time update we have

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, 0)$$

- its corresponding covariance matrix

$$\hat{P}_{k|k-1} = F_k \hat{P}_{k-1|k-1} F_k^T + U_k Q_{k-1} U_k^T$$

- the Kalman gain matrix, K_k is defined in the measurement equation as

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - h(\hat{x}_{k|k-1}, 0))$$

Extended Kalman Filter

- corresponding covariance matrix

$$P_{k|k} = (I - K_k H_k) P_{k|k-1}$$

- as in the Kalman filter, the optimal Kalman gain that minimizes the mean square error over all linear estimators is given by

$$K_k = P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + V_k R_k V_k^\top)^{-1}$$

Extended Kalman Filter

by Ali Hirs

Example (Parameter estimation of Heston model via EKF)

we set up parameter estimation of Heston model via EKF. In Heston, the underlying process follows:

$$\begin{aligned} dS_t &= \mu S_t dt + \sqrt{v_t} S_t dW_t^1 \\ dv_t &= \kappa(\theta - v_t)dt + \lambda\sqrt{v_t}dW_t^2 \end{aligned}$$

where the two Brownian components W_t^1 and W_t^2 are correlated with rate ρ under physical measure. Define $y_t = \ln(S_t)$ and using Itô's lemma we can write it as

$$\begin{aligned} dy_t &= \left(\mu - \frac{1}{2}v_t\right)dt + \sqrt{v_t}dW_t^1 \\ dv_t &= \kappa(\theta - v_t)dt + \lambda\sqrt{v_t}dW_t^2 \end{aligned}$$

For

Extended Kalman Filter

for Heston, the state equation is

$$\begin{aligned} x_k = f(x_{k-1}, u_k) &= \begin{pmatrix} y_k \\ v_k \end{pmatrix} \\ &= \begin{pmatrix} y_{k-1} + (\mu - \frac{1}{2}v_{k-1})\Delta t + \sqrt{v_{k-1}}\sqrt{\Delta t}Z_k^1 \\ v_{k-1} + \kappa(\theta - v_{k-1})\Delta t + \lambda\sqrt{v_{k-1}}\sqrt{\Delta t}Z_k^2 \end{pmatrix} \end{aligned}$$

with system noise

$$u_k = \begin{pmatrix} Z_k^1 \\ Z_k^2 \end{pmatrix}$$

and the covariance matrix

$$Q_k = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

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Extended Kalman Filter

it is easy to see that in the extended Kalman filter, F_k and U_k for Heston stochastic volatility are

$$F_k = \begin{pmatrix} 1 & -\frac{1}{2}\Delta t \\ 0 & 1 - \kappa\Delta t \end{pmatrix}$$

and

$$U_k = \begin{pmatrix} \sqrt{v_{k-1}}\sqrt{\Delta t} & 0 \\ 0 & \lambda\sqrt{v_{k-1}}\sqrt{\Delta t} \end{pmatrix}$$

assume measurement equation $y_k = \ln(S_k)$, which implies

$$H_k = [1 \ 0]$$

Extended Kalman Filter

for a given set of parameters $\Theta = \{\kappa, \theta, \lambda, \rho, v_0\}$, we would minimize the following summation as our objective function to obtain the optimal parameter set for the model⁵

$$\sum_{i=1}^N \left(\ln(S_k) + \frac{e_k^2}{S_k} \right)$$

where

$$S_k = H_k P_{k|k-1} H_k^T + V_k R_k V_k^T$$

and

$$e_k = z_k - h(\hat{x}_{k|k-1}, 0)$$

as mentioned earlier, minimization can be performed via the Nelder-Mead optimization algorithm

⁵referring to Equation (1) and note that for this case S_k is scalar

Extended Kalman Filter

Algorithm 1: pseudo-code for implementation

Input: $v = v_0 = 0.005$, $\log \ell_k = 0$, $H = [1 \ 0]$

$$F = \begin{pmatrix} 1 & -\frac{1}{2}\Delta t \\ 0 & 1 - \kappa\Delta t \end{pmatrix} \quad Q = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \quad P = \begin{pmatrix} 0.01 & 0 \\ 0 & 0.01 \end{pmatrix}$$

```

1  for k=2:n do
2
    $U = \begin{pmatrix} \sqrt{v}\sqrt{\Delta t} & 0 \\ 0 & \lambda\sqrt{v}\sqrt{\Delta t} \end{pmatrix}$ 
3   $y_{k-1} = \ln(S_{k-1})$ 
4
    $x_1 = \begin{pmatrix} y_{k-1} + (\mu - \frac{1}{2}v)\Delta t \\ v + \kappa(\theta - v)\Delta t \end{pmatrix}$ 
5   $P_1 = FPF^T + UQU^T$ 
6   $S = HP_1H^T$  (scalar)
7   $K = P_1H^T / S$  (scalar)
8   $v_1 = x_1[2]$  second element of vector  $x_1$ 
9   $e = (y_k - y_{k-1}) - (\mu - \frac{1}{2}v_1)\Delta t$ 
10  $\log \ell_k = \log \ell_k + \log(S) + e^2 / S$ 
11  $x = x_1 + K \times e$ 
12  $v = x[2]$  second element of vector  $x$ 
13  $P = (I - KH) \times P_1$  updating  $P$ 
14 end
  
```

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Unscented Kalman Filter

- when state transition and observation models, i.e. the predict and update functions f and h are highly non-linear, the extended Kalman filter can give particularly poor performance
- this is because only the mean is propagated through the non-linearity
- the unscented Kalman filter (UKF) uses a deterministic sampling technique known as the unscented transform to pick a minimal set of sample points called sigma points around the mean
- these sigma points are then propagated through the non-linear functions, and the covariance of the estimate is then recovered
- the result is a filter which more accurately captures the true mean and covariance

Unscented Kalman Filter

- this can be verified using Monte Carlo sampling or through a Taylor series expansion of the posterior statistics
- in addition, this technique removes the requirement to explicitly calculate Jacobians, which for complex functions can be a difficult task in itself (i.e., requiring complicated derivatives if done analytically or being computationally costly if done numerically)

Predict

- as with the EKF, the UKF prediction can be used independently from the UKF update, in combination with a linear (or indeed EKF) update, or vice versa
- the estimated state and covariance are augmented with the mean and covariance of the process noise

$$\begin{aligned}
 x_{k-1|k-1}^a &= \begin{bmatrix} \hat{x}_{k-1|k-1}^\top \\ \mathbb{E}[w_k^\top] \end{bmatrix} \\
 P_{k-1|k-1}^a &= \begin{bmatrix} P_{k-1|k-1} & 0 \\ 0 & Q_k \end{bmatrix}
 \end{aligned}$$

- a set of $2L + 1$ sigma points is derived from the augmented state and covariance where L is the dimension of the augmented state

Predict

$$\chi_{k-1|k-1}^0 = x_{k-1|k-1}^a$$

$$\chi_{k-1|k-1}^i = x_{k-1|k-1}^a + \left(\sqrt{(L + \lambda) P_{k-1|k-1}^a} \right)_i \quad i = 1, \dots, L$$

$$\chi_{k-1|k-1}^i = x_{k-1|k-1}^a - \left(\sqrt{(L + \lambda) P_{k-1|k-1}^a} \right)_{i-L} \quad i = L+1, \dots, 2L$$

where

$$\left(\sqrt{(L + \lambda) P_{k-1|k-1}^a} \right)_i$$

is the i -th column of the matrix square root of

$$(L + \lambda) P_{k-1|k-1}^a$$

the matrix square root should be calculated using numerically efficient and stable methods such as the Cholesky decomposition.

Predict

the sigma points are propagated through the transition function f .

$$\chi_{k|k-1}^i = f(\chi_{k-1|k-1}^i) \quad i = 0, \dots, 2L$$

the weighted sigma points are recombined to produce the predicted state and covariance.

$$\hat{x}_{k|k-1} = \sum_{i=0}^{2L} W_i^{(m)} \chi_{k|k-1}^i$$

$$P_{k|k-1} = \sum_{i=0}^{2L} W_i^{(c)} [\chi_{k|k-1}^i - \hat{x}_{k|k-1}][\chi_{k|k-1}^i - \hat{x}_{k|k-1}]$$

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Predict

weights for the state and covariance are given by

$$\begin{aligned}
 W_0^{(m)} &= \frac{\lambda}{L + \lambda} \\
 W_0^{(c)} &= \frac{\lambda}{L + \lambda} + (1 - \alpha^2 + \beta) \\
 W_i^{(m)} &= \frac{1}{2(L + \lambda)} \quad i = 1, \dots, 2L \\
 W_i^{(c)} &= \frac{1}{2(L + \lambda)} \quad i = 1, \dots, 2L
 \end{aligned}$$

where

$$\lambda = \alpha^2(L + \kappa) - L$$

Update

the predicted state and covariance are augmented as before, except now with the mean and covariance of the measurement noise.

$$x_{k|k-1}^a = \begin{bmatrix} \hat{x}_{k|k-1}^\top \\ \mathbb{E}[v_k^\top] \end{bmatrix}$$

$$P_{k|k-1}^a = \begin{bmatrix} P_{k|k-1} & 0 \\ 0 & R_k \end{bmatrix}$$

as before, a set of $2L + 1$ sigma points is derived from the augmented state and covariance where L is the dimension of the augmented state.

Update

$$\chi_{k|k-1}^0 = x_{k|k-1}^a$$

$$\chi_{k|k-1}^i = x_{k|k-1}^a + \left(\sqrt{(L + \lambda) P_{k|k-1}^a} \right)_i \quad i = 1, \dots, L$$

$$\chi_{k|k-1}^i = x_{k|k-1}^a - \left(\sqrt{(L + \lambda) P_{k|k-1}^a} \right)_{i-L} \quad i = L + 1, \dots, 2L$$

alternatively, if the UKF prediction has been used, the sigma points themselves can be augmented along the following lines:

$$\chi_{k|k-1} = [\chi_{k|k-1}^\top \mathbb{E}[v_k^\top]^\top \pm \sqrt{(L + \lambda) R_k^a}]$$

where

$$R_k^a = \begin{bmatrix} 0 & 0 \\ 0 & R_k \end{bmatrix}$$

Update

the sigma points are projected through the observation function h .

$$\gamma_k^i = h(\chi_{k|k-1}^i) \quad i = 0, \dots, L$$

the weighted sigma points are recombined to produce the predicted measurement and predicted measurement covariance.

$$\hat{z}_k = \sum_{i=0}^{2L} W_i^{(m)} \gamma_k^i$$

$$P_{z_k z_k} = \sum_{i=0}^{2L} W_i^{(c)} [\gamma_k^i - \hat{z}_k][\gamma_k^i - \hat{z}_k]^\top$$

the state-measurement cross-covariance matrix

$$P_{x_k z_k} = \sum_{i=0}^{2L} W_i^{(c)} [\chi_{k|k-1}^i - \hat{x}_{k|k-1}][\gamma_k^i - \hat{z}_k]^\top$$

is used to compute the UKF Kalman gain

Update

$$K_k = P_{x_k z_k} P_{z_k z_k}^{-1}$$

as with the Kalman filter, the updated state is the predicted state plus the innovation weighted by the Kalman gain

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - \hat{z}_k)$$

and the updated covariance is the predicted covariance minus the predicted measurement covariance, weighted by the Kalman gain.

$$P_{k|k} = P_{k|k-1} - K_k(P_{z_k z_k} K_k^T)$$

Implementation of Unscented Kalman Filter (UKF)

1 Initialize

$$\begin{aligned}\hat{x}_0 &= \mathbb{E}(x_0) \\ P_0 &= \mathbb{E}[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^\top]\end{aligned}$$

2 Calculate sigma points

$$\mathbf{x}_{k-1} = \begin{bmatrix} \hat{x}_{k-1} & \hat{x}_{k-1} + \gamma \sqrt{P_{k-1}} & \hat{x}_{k-1} - \gamma \sqrt{P_{k-1}} \end{bmatrix}$$

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Implementation of Unscented Kalman Filter (UKF)

3 Time update

$$\mathbf{x}_{k|k-1}^* = F[\mathbf{x}_{k-1}, u_{k-1}]$$

$$\hat{\mathbf{x}}_k^- = \sum_{i=0}^{2L} W_i^{(m)} \mathbf{x}_{i,k|k-1}^*$$

$$P_k^- = \sum_{i=0}^{2L} W_i^{(c)} [\mathbf{x}_{i,k|k-1}^* - \hat{\mathbf{x}}_k^-][\mathbf{x}_{i,k|k-1}^* - \hat{\mathbf{x}}_k^-]^\top + Q$$

$$\mathbf{x}_{k|k-1} = \begin{bmatrix} \hat{\mathbf{x}}_k^- & \hat{\mathbf{x}}_k^- + \gamma \sqrt{P_k^-} & \hat{\mathbf{x}}_k^- - \gamma \sqrt{P_k^-} \end{bmatrix}$$

$$\mathbf{y}_{k|k-1} = H[\mathbf{x}_{k|k-1}]$$

$$\hat{\mathbf{y}}_k^- = \sum_{i=0}^{2L} W_i^{(m)} \mathbf{y}_{i,k|k-1}$$

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Implementation of Unscented Kalman Filter (UKF)

4 measurement update equations

$$P_{\hat{y}_k \hat{y}_k} = \sum_{i=0}^{2L} W_i^{(c)} [\mathbf{y}_{i,k|k-1} - \hat{y}_k^-] [\mathbf{y}_{i,k|k-1} - \hat{y}_k^-]^\top + R$$

$$P_{x_k y_k} = \sum_{i=0}^{2L} W_i^{(c)} [\mathbf{x}_{i,k|k-1} - \hat{x}_k^-] [\mathbf{y}_{i,k|k-1} - \hat{y}_k^-]^\top$$

$$K_k = P_{x_k y_k} P_{\hat{y}_k \hat{y}_k}^{-1}$$

$$\hat{x}_k = \hat{x}_k^- + K_k (y_k - \hat{y}_k^-)$$

$$P_k = P_k^- - K_k P_{\hat{y}_k \hat{y}_k} K_k^\top$$

where Q and R are the process noise covariance and measurement noise covariance matrices, respectively

Implementation of Unscented Kalman Filter (UKF)

- for discrete time and continuous time models, we can get the analytical form of R^v , but we do not have the formula for R
- we can get this matrix through a parameter estimation process, or for simplicity we assume $R = \delta I$, where δ is a small scaling value and I is the identity matrix

Some comments (1 of 2)

- (1) the square root of P is defined by $SS^T = P$, which requires a Cholesky factorization.
- (2) to get the square root of P , P must be positive semi-definite
 - ▷ in implementation, if P has a negative eigenvalue we modify it by adding a diagonal matrix with very small entries on the diagonal, that is, $P = P + \delta I$
 - ▷ we keep doing it until all eigenvalues of P become non-negative
 - ▷ given a proper model parameter set, the filter convergence would be fast, P stays positive semi-definite, and no modification is needed

Some comments (2 of 2)

- (3) to calculate the Kalman gain, the inverse matrix of $P_{\hat{y}_k \hat{y}_k}$ is needed
- ▷ if $P_{\hat{y}_k \hat{y}_k}$ is likely singular, the inverse matrix would have extreme values, consequently the new sigma points would be in an unreasonably large range
 - ▷ so adding a proper R_n is a must
- (4) we use $\mathbf{x}_{k|k-1} = \mathbb{E}(\mathbf{x}_{k-1})$ to calculate the one step forward sigma points and $\text{cov}(\mathbf{x}_{k-1})$ is the process noise covariance Q

Square Root Unscented Kalman Filter (SR UKF)

- implementation of Cholesky factorization in UKF requires $O(L^3/6)$ computations where L is the state dimension, a square root unscented Kalman filter can reduce it to $O(L^2)$; and speed up the filtering process especially for the system that has a large state dimension
- here are the implementation steps:

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Implementation of SR UKF

1 initialize

$$\hat{x}_0 = \mathbb{E}(x_0)$$

$$S_0 = \text{chol} \left(\mathbb{E}[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T] \right)$$

$$x_{t0} = -B^{-1}b_\gamma$$

$$v_{t0} = \theta$$

$$\hat{x}_0 = \begin{bmatrix} x_{t0} \\ v_{t0} \end{bmatrix}$$

and $S_0 = \text{chol}(R_v)$ or identity matrix for simplicity.

2 calculate sigma points

$$\mathbf{x}_{k-1} = [\hat{x}_{k-1} \quad \hat{x}_{k-1} + \gamma S_k \quad \hat{x}_{k-1} - \gamma S_k]$$

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Implementation of SR UKF

3 time update

$$\mathbf{x}_{k|k-1}^* = F[\mathbf{x}_{k-1}, u_{k-1}]$$

$$\hat{\mathbf{x}}_k^- = \sum_{i=0}^{2L} W_i^{(m)} \mathbf{x}_{i,k|k-1}^*$$

$$S_k^- = qr\{[\sqrt{W_1^{(c)}} [\mathbf{x}_{1:2L,k|k-1}^* - \hat{\mathbf{x}}_k^-] \quad \sqrt{R^v}]\}$$

$$S_k^- = cholupdate\{S_k^-, \mathbf{x}_{0,k}^* - \hat{\mathbf{x}}_k^-, W_0^{(c)}\}$$

$$\mathbf{x}_{k|k-1} = \begin{bmatrix} \hat{\mathbf{x}}_k^- & \hat{\mathbf{x}}_k^- + \gamma S_k^- & \hat{\mathbf{x}}_k^- - \gamma S_k^- \end{bmatrix}$$

$$\mathbf{y}_{k|k-1} = H[\mathbf{x}_{k|k-1}]$$

$$\hat{\mathbf{y}}_k^- = \sum_{i=0}^{2L} W_i^{(m)} \mathbf{y}_{i,k|k-1}$$

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Implementation of SR UKF

④ measurement update equations

$$S_{\hat{y}_k} = qr\{[\sqrt{W_1^{(c)}}[\mathbf{y}_{1:2L,k} - \hat{y}_k] \quad \sqrt{R_k^n}]\}$$

$$S_{\hat{y}_k} = cholupdate\{S_{\hat{y}_k}, \mathbf{y}_{0,k} - \hat{y}_k, W_0^{(c)}\}$$

$$P_{x_k y_k} = \sum_{i=0}^{2L} W_i^{(c)} [\mathbf{x}_{i,k|k-1} - \hat{x}_k^-][\mathbf{y}_{i,k|k-1} - \hat{y}_k^-]^\top$$

$$K_k = (P_{x_k y_k} / S_{\hat{y}_k}^\top) / S_{\hat{y}_k}$$

$$\hat{x}_k = \hat{x}_k^- + K_k(y_k - \hat{y}_k^-)$$

$$U = K_k S_{y_k}$$

$$S_k = cholupdate\{S_k^-, U, -1\}$$

the variables used are defined as

$$W_0^m = \lambda / (L + \lambda)$$

$$W_0^{(c)} = \lambda / (L + \lambda) + (1 - \alpha^2 + \beta)$$

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Some comments

- (1) QR decomposition – we define $A^T = QR$, where $A \in \mathbb{R}^{L \times N}$, and $N \geq L$. The Q is an orthogonal matrix, R is an upper triangular matrix, and the $\text{qr}\{\cdot\}$ in the algorithm denotes the upper triangular part of R
- (2) Cholesky factor updating – $\text{cholupdate}(S, u, \pm v)$ where $S = \text{chol}(A)$ is the original Cholesky factorization of A , returns the upper triangular Cholesky factor of $A \pm \sqrt{v}uu^T$, where u is a column vector of appropriate length. If u is a matrix not a vector, then the result is M consecutive updates of the Cholesky factor use the M column of u . This algorithm is $O(L^2)$ per update
- (3) Given the same initial value and model parameters, the filtering result of UKF and SR-UKF should be the same. One exception is when P is not positive semi-definite during UKF, and has been added an I

Particle Filter

- a more recent alternative for filtering nonlinear processes is provided by a collection of techniques known as *particle filtering*
- particle filters, also known as *sequential Monte Carlo*, approximate the continuous density function $p(x_t|y_t)$ by a discrete density function, effectively a histogram via simulation
- the idea, based on importance sampling, is to use Monte Carlo simulation to replace the Gaussian approximation for $p(x_t|y_t)$ that was used in the Kalman filter or the extended Kalman filter
- thus a better estimate of the parameters from fundamentally nonlinear processes might be obtained

Particle Filter

- like all sampling based approaches we generate a set of samples that approximate the distribution function $p(x_t|y_t)$
- for N sample points we have $\{x_t^{(i)}, w_t^{(i)}\}_{i=1}^N$ where $x_t^{(i)}$ are support points to the discrete distribution called particles and $w_t^{(i)}$ is weight associated to support point $x_t^{(i)}$ and obviously sum of the weights are one
- the expectation with respect to the filtering is approximated by

$$\mathbb{E}(f(x_t)) = \int f(x_t)p(x_t|y_t)dx_t \approx \sum_{i=1}^N w_t^{(i)} f(x_t^{(i)})$$

Particle Filter

- like other filtering techniques, the first step in the particle filtering algorithm is initialization. The choice of a proper initial state value plays a more crucial role in particle filtering than in the unscented Kalman filter
- a bad initial value could make the sigma points (the particles) far from the actual value, and the probability of a particle to the actual state can be very small; therefore the converge process could take a long time
- one solution of finding a proper initial state value is using UKF for the prior state value. The UKF method converges quickly to the true state value; then one would switch to a particle filter for more accurate estimation

Particle Filter

- for a finite set of particles, the algorithm performance depends on the choice of the proposal distribution, $\pi(x_k|x_{0:k-1}, y_{0:k})$
- the optimal proposal distribution is given by the target distribution

$$\pi(x_k|x_{0:k-1}, y_{0:k}) = p(x_k|x_{k-1}, y_k)$$

- however, the transition prior is often used as the importance function, since it is easier to draw particles and perform subsequent importance weight calculations as

$$\pi(x_k|x_{0:k-1}, y_{0:k}) = p(x_k|x_{k-1})$$

Sequential Importance Sampling (SIS) Particle Filtering

the sequential importance sampling (SIS) algorithm is as follows:

1. simulate the state from the prior (or another proposal distribution) that is drawing N samples according to the model

$$x_k^{(i)} = f(x_{k-1}^{(i)}, u_{k-1}^{(i)}), \quad i = 1, 2, \dots, N$$

2. associate to each simulated point a weight. That is done by updating the importance weights

$$w_k^{(i)} = w_{k-1}^{(i)} \frac{p(z_k | x_k^{(i)}) p(x_k^{(i)} | x_{k-1}^{(i)})}{\pi(x_k^{(i)} | x_{k-1}^{(i)}, z_k)}$$

where $p(z_k | x_k^{(i)})$ is probability, $p(x_k^{(i)} | x_{k-1}^{(i)})$ is transition probability, and $\pi(x_k^{(i)} | x_{k-1}^{(i)}, z_k)$ is the proposal distribution

Sequential Importance Sampling (SIS) Particle Filtering

- if we assume $\pi(x_k^{(i)} | x_{k-1}^{(i)}, z_k) = p(x_k^{(i)} | x_{k-1}^{(i)})$, the expression can be simplified as

$$w_k^{(i)} = w_{k-1}^{(i)} p(z_k | x_k^{(i)})$$

3. Normalize the weights $\tilde{w}_k(x_k^{(i)}) = \frac{w_k^{(i)}}{\sum_j w_k^{(j)}}$

then the best approximation of x_k is its conditional expectation given $z_{1:k}$, that is,

$$\mathbb{E}(x_k | z_{1:k}) \approx \sum_{i=1}^{N_{sims}} \tilde{w}_k(x_k^{(i)}) x_k^{(i)}$$

Sampling Importance Resampling (SIR) Particle Filtering

- the variance of the weights increases over time in SIS, so the algorithm will diverge
- this is known as the **degeneracy problem** in sampling importance sampling
- a solution is proposed by Arulampalam et al. that has to do with resampling
- that is, regenerate particles with higher weight and eliminate those with lower weight

Sampling Importance Resampling (SIR) Particle Filtering

- the resampling algorithm is done as follows: compare the cumulative distribution function (CDF) created from the normalized weights with a CDF constructed from a uniformly simulated number $\mathcal{U}[0, 1]$
- at time step k and for $j = 1, \dots, N_{sims}$, if

$$\frac{1}{N_{sims}}(\mathcal{U}[0, 1] + j - 1) \geq \sum_{l=1}^i \tilde{w}_k(x_k^{(l)})$$

then increment and *skip* i , otherwise take $x_k^{(i)}$ and set its weight to $\frac{1}{N_{sims}}$

Sampling Importance Resampling (SIR) Particle Filtering

here is the pseudo-code for the resampling algorithm:

for $j = 1, \dots, N_{sims}$

$$c(j) = \sum_{l=1}^j \tilde{w}_k(x_k^{(l)})$$

end for

$i = 1;$

for $j = 1, \dots, N_{sims}$

$$u(j) = \frac{1}{N_{sims}} (\mathcal{U}[0,1] + j - 1)$$

while $(u(j) > c(i))$

$i = i + 1;$

end while

$$\tilde{x}_k^{(j)} = x_k^{(i)}$$

$$\tilde{w}_k^j = \frac{1}{N_{sims}}$$

end for

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Sampling Importance Resampling (SIR) Particle Filtering

the sampling importance resampling (SIR) algorithm is as follows:

1. simulate the state from the prior (or another proposal distribution)
2. associate to each simulated point a weight equal to the conditional likelihood density $w_k^{(i)} = w_{k-1}^{(i)} p(z_k | x_k^{(i)})$
3. normalize the weights $\tilde{w}_k(x_k^{(i)}) = \frac{w_k^{(i)}}{\sum_j w_k^{(j)}}$
4. resample according the pseudo-code explained earlier

Example (1 of 9)

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Example (Particle filter for Heston stochastic volatility model)

For the Heston stochastic volatility model we use the following densities:

$$\pi(x_k^{(i)} | x_{k-1}^{(i)}, y_k) = n\left(x_k^{(i)}, x_{k-1}^{(i)}, \sqrt{P_k^{(i)}}\right)$$

$$p(y_k | x_k^{(i)}) = n\left(y_k, y_{k-1} + \left(\mu - \frac{1}{2}x_k^{(i)}\right)\Delta t, \sqrt{x_k^{(i)}\Delta t}\right)$$

$$p(x_k^{(i)} | x_{k-1}^{(i)}) = n\left(x_k^{(i)}, x_{k-1}^{(i)} + \delta\Delta t + \rho\lambda(y_{k-1} - y_{k-2}), \lambda\sqrt{1-\rho^2}\sqrt{x_{k-1}^{(i)}\Delta t}\right)$$

where

$$\delta = \kappa\theta - \rho\lambda\mu - \left(\kappa - \frac{1}{2}\rho\lambda\right)x_{k-1}^{(i)}$$

$$n(x, m, s) = \frac{1}{\sqrt{2\pi}s} \exp\left(-\frac{(x - m)^2}{2s^2}\right)$$

For

Example (2 of 9)

- use two different approaches for the Heston stochastic volatility model
- one approach is for parameter estimation and the other one is for prediction
- in the first approach we follow the work done by Aihara et al. for filtering and prediction and in the second approach we follow the work done by Mimouni for parameter estimation
- implementation details of the first approach for the Heston stochastic volatility model (Heston Model-1) is as follows:
- start from the Heston SDE

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dw_t^1 \quad (2)$$

$$dv_t = \kappa(\theta - v_t)dt + \lambda\sqrt{v_t}dw_t^2 \quad (3)$$

Example (3 of 9)

- define $y_t = \log S_t$ using Itô's lemma we can rewrite (2) as follows:

$$dy_t = \left(\mu - \frac{1}{2}v_t\right)dt + \sqrt{v_t}dw_t^1 \quad (4)$$

- knowing the correlation between dw_t^1 and dw_t^2 is ρ , $\langle dw_t^1, dw_t^2 \rangle = \rho dt$, we can write

$$dw_t^2 = \rho dw_t^1 + \sqrt{1 - \rho^2} dz_t \quad (5)$$

and substituting it into (3) we get

$$dv_t = \kappa(\theta - v_t)dt + \lambda\sqrt{v_t}(\rho dw_t^1 + \sqrt{1 - \rho^2} dz_t) \quad (6)$$

Example (4 of 9)

from (4) we can write

$$dw_t^1 = \frac{dy_t - (\mu - \frac{1}{2}v_t)dt}{\sqrt{v_t}} \quad (7)$$

by substituting (7) in (6) we obtain

$$\begin{aligned} dv_t &= \kappa(\theta - v_t)dt + \lambda\sqrt{v_t}\left(\rho\frac{dy_t - (\mu - \frac{1}{2}v_t)dt}{\sqrt{v_t}} + \sqrt{1 - \rho^2}dz_t\right) \\ &= \kappa(\theta - v_t)dt + \lambda\rho(dy_t - (\mu - \frac{1}{2}v_t)dt) + \lambda\sqrt{v_t}\sqrt{1 - \rho^2}dz_t \end{aligned} \quad (8)$$

rewriting the discrete version of the state equation (8)

$$v_t = v_{t-1} + \kappa(\theta - v_{t-1})\Delta t + \lambda\rho(y_t - y_{t-1} - (\mu - \frac{1}{2}v_{t-1})\Delta t) + \lambda\sqrt{v_{t-1}(1 - \rho^2)}\Delta t z_1 \quad (9)$$

where $z_1 \sim \mathcal{N}(0, 1)$.

Example (5 of 9)

now we use explicit-implicit discretization to discretize (4) as follows:

$$y_t = y_{t-1} + (\mu - \frac{1}{2}v_t)\Delta t + \sqrt{v_{t-1}}\sqrt{\Delta t} z_2 \quad (10)$$

and plug (10) into (9) to get

$$\begin{aligned} v_t = & v_{t-1} + \kappa(\theta - v_{t-1})\Delta t + \lambda\rho((\mu - \frac{1}{2}v_t)\Delta t - (\mu - \frac{1}{2}v_{t-1})\Delta t) \\ & + \lambda\sqrt{v_{t-1}(1 - \rho^2)}\Delta z_2 + \lambda\rho\sqrt{v_{t-1}}\sqrt{\Delta t}z_1 \end{aligned} \quad (11)$$

notice that if we did not use explicit-implicit discretization the terms in (11) would have canceled each other. By doing this we make the scheme stable.

Example (6 of 9)

from (9) we can construct the optimal proposal (importance) function, $\pi(v_t|v_{t-1}, y_t)$, that is,

$$\pi(v_t|v_{t-1}, y_t) = n(v_t, m_t, \sigma_t) \quad (12)$$

where

$$m_t = v_{t-1} + \kappa(\theta - v_{t-1})\Delta t + \lambda\rho(y_t - y_{t-1} - (\mu - \frac{1}{2}v_{t-1})\Delta t)$$

$$\sigma_t = \lambda\sqrt{v_{t-1}(1 - \rho^2)\Delta t}$$

$$n(x, m, s) = \frac{1}{\sqrt{2\pi}s} \exp\left(-\frac{(x - m)^2}{2s^2}\right)$$

Example (7 of 9)

from (10), we have the likelihood function $p(y_t | v_t, v_{t-1}, y_{t-1})$

$$p(y_t | v_t, v_{t-1}, y_{t-1}) = n(y_t, m_L, \sigma_L)$$

where

$$m_L = y_{t-1} + \left(\mu - \frac{1}{2}v_t\right)\Delta t$$

$$\sigma_L = \sqrt{v_{t-1}}\Delta t$$

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Example (8 of 9)

from (11) we have the transition density $p(v_t | v_{t-1})$

$$p(v_t | v_{t-1}) = n(v_t, m_T, \sigma_T) \quad (13)$$

where

$$m_T = \left(1 + \frac{1}{2}\lambda\rho\Delta t\right)^{-1}(v_{t-1} + \kappa(\theta - v_{t-1})\Delta t + \frac{1}{2}\lambda\rho v_{t-1}\Delta t)$$

$$\sigma_T = \left(1 + \frac{1}{2}\lambda\rho\Delta t\right)^{-1}\lambda\sqrt{v_{t-1}\Delta t}$$

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Example (9 of 9)

then weights are updated as follows: initial weights are all equal to $\frac{1}{N}$, then

$$w_t^i = w_{t-1}^i \frac{p(y_t | v_t, v_{t-1}, y_{t-1}) p(v_t | v_{t-1})}{\pi(v_t | v_{t-1}, y_t)}$$

then the likelihood is approximated by $\sum_{t=1}^T (\log \sum_{i=1}^N w_t^i)$ where T is number of days and N is number of particles used.

Implementation of the prediction step (1 of 3)

$$\begin{aligned}
 p(y_t | y_{1:t-1}) &= \int p(y_t, x_{t-1} | y_{1:t-1}) dx_{t-1} \\
 &= \int p(y_t | x_{t-1}) p(x_{t-1} | y_{1:t-1}) dx_{t-1} \\
 &= \int \int p(y_t, x_t | x_{t-1}) p(x_{t-1} | y_{1:t-1}) dx_t dx_{t-1} \\
 &= \int \int p(y_t | x_t) p(x_t | x_{t-1}) p(x_{t-1} | y_{t-1}) dx_t dx_{t-1}
 \end{aligned}$$

the above equation suggests the following approximation for predictive density $p(y_t | y_{1:t-1})$.

Implementation of the prediction step (2 of 3)

suppose $x_{t-1}^{(i)}$ is the i -th sample from $p(x_{t-1}|y_{1:t-1})$ and $x_{t-1}^{i,j}$ is the j -th sample from $p(x_t|x_{t-1}^{(i)})$, then we can estimate each $p(y_t|y_{1:t-1})$ via

$$p(y_t|y_{1:t-1}) = \frac{1}{N \times M} \sum_{i=1}^N \sum_{j=1}^M p(y_t|x_t^{i,j})$$

we implement the prediction as follows: given time $t-1$ (today), particles $v_{t-1}^{(i)}$ is as approximation for posterior samples $p(v_{t-1}|y_{1:t-1})$, doing one step propagation of the state equation to generate v_t^i via the following equation

$$v_t^{(i)} = v_{t-1}^{(i)} + \kappa(\theta - v_{t-1}^{(i)})\Delta t$$

given $v_t^{(i)}$, we predict y_t^i using the following equation:

$$y_t^i = y_{t-1} + (\mu - \frac{1}{2}v_t^i)\Delta t + \sqrt{v_{t-1}^i}\Delta w^1(t)$$

Implementation of the prediction step (3 of 3)

then given $y_t^k, k = 1, \dots, N$, we calculate the predictive density of y_t^k by

$$p(y_t^k | y_{1:t-1}) = \frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M p(y_t^k | x_t^{i,j})$$

simplify it by using $M=1$, then $p(y_t^k | y_{1:t-1}) = \frac{1}{N} \sum_{i=1}^N p(y_t^k | x_t^i)$,
then the prediction of log stock price at time t is given by

$$y_t^{pred} = \sum_{k=1}^N p(y_t^k | y_{1:t-1}) y_t^k$$